

OmiCoreTumor: an open, molecularly validated model for mapping tumour regions in colorectal cancer H&E

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Preprint. Research use only — not a medical device and not validated for clinical use.

Abstract

Selecting tumour regions on haematoxylin and eosin (H&E) sections is a routine prerequisite for spatial-omics experiments, yet it is usually done by hand and is rarely reproducible between operators. We present OmiCoreTumor (omicoretumor-crc-he-v0.1), an openly licensed model that maps tumour-enriched regions in colorectal H&E and exports them directly as QuPath-readable annotations.

The model is an ensemble of three convolutional classifiers trained on 100,000 openly licensed tissue tiles, combined with Macenko stain normalisation applied at inference. We show that the dominant obstacle to deployment is not discrimination but stain-domain shift: a single model retained an AUROC of 0.955 on unseen slides while its recall at the conventional 0.5 threshold fell to 0.48. Training without stain normalisation raised cross-cohort tumour AUROC from 0.836 to 0.992, and normalising at inference reduced false-positive area on tumour-free tissue by up to 27-fold. Combining both recovers recall.

On five 10x Visium HD colorectal sections that are independent of all training data, the released model called 36% of tissue as tumour in three carcinoma sections (27.7%–48.5%) and under 1.8% in two matched normal controls, exporting no tumour regions at all from either control. On the one section with matched transcriptomics, agreement with Visium HD cell-type labels reached AUROC 0.9855 (precision 0.935, sensitivity 0.921, specificity 0.962). Because the image model never observes gene expression, this is orthogonal evidence rather than a comparison between two morphology-derived annotations.

The model, weights, training code and evaluation are released under Apache-2.0 as a pip-installable package, so that any group can map tumour regions on their own colorectal sections with three lines of Python. We are explicit about what the evidence does and does not support: the model localises tumour-enriched regions at 112 um resolution across five sections from one cohort; it does not identify malignant cells, and it has not been shown to generalise across scanners, histological variants or difficult non-neoplastic tissue.

Availability

```
pip install omicoretumor
github.com/OmiCore/omicoretumor    Apache-2.0 (code)
Model: omicoretumor-crc-he-v0.1
```

Introduction

Spatial transcriptomics experiments on solid tumours almost always begin with a morphological question: which part of this section is tumour? The answer determines which regions are profiled, how compartments are compared, and how results are interpreted. In practice the region is drawn by hand, which is slow, operator-dependent and difficult to report reproducibly in a methods section.

Tile-level tissue classifiers for colorectal H&E have existed since the release of the NCT-CRC-HE-100K dataset, and reach very high accuracy on held-out tiles from the same cohort. That accuracy has not translated into tools that groups routinely apply to their own slides. We found the reason is mundane but decisive: the probability calibration of these models does not survive a change of stain and scanner, so a model that still ranks tiles correctly will silently miss most of the tumour when its usual threshold is applied to a new laboratory's sections.

We therefore treated robustness, not headline accuracy, as the design target, and validated the result against a modality the model cannot see.

Results

Stain-domain shift, not discrimination, limits transfer (Fig. 1, Fig. 2). Three classifiers were trained on 100,000 tiles spanning nine tissue classes. On CRC-VAL-HE-7K, a held-out cohort processed identically to the training data, the normalisation-trained model reached 0.970 accuracy and 0.999 tumour AUROC. On Kather-2016, a different cohort and scanner, the same model fell to 0.836 tumour AUROC, whereas an identically trained model that had seen non-normalised tiles reached 0.992 (Fig. 2c). Measuring the colour statistics of real target slides showed why: they sit a mean RGB distance of 56 from the training domain, which Macenko normalisation reduces to 28 (Fig. 2a,b).

The practical consequence is a calibration failure rather than a ranking failure. A single model applied to raw target slides kept essentially unchanged discrimination (AUROC 0.955) while its recall at the conventional 0.5 threshold fell to 0.48 (Fig. 2e). The two molecular classes remained well separated, but the probabilities had compressed toward zero (Fig. 2d). Any deployment that inherits a threshold from the training domain will therefore appear highly precise while discarding much of the tumour. Combining non-normalised training with inference-time normalisation addresses both halves of the problem, and this variant ranked first of seven on the independent section-level criterion (Fig. 1d).

Performance on tumour and control sections (Fig. 3). Applied to five Visium HD colorectal sections, the released model called 32.3%, 48.5% and 27.7% of tissue as tumour in the three carcinoma sections, against 0.08% and 1.82% in the two matched normal controls. After the minimum-area filter used for annotation export, neither control produced a single tumour region. The predicted tissue composition is biologically coherent (Fig. 3g): tumour epithelium and desmoplastic stroma dominate the carcinoma sections, while normal mucosa, smooth muscle and mucus dominate the controls.

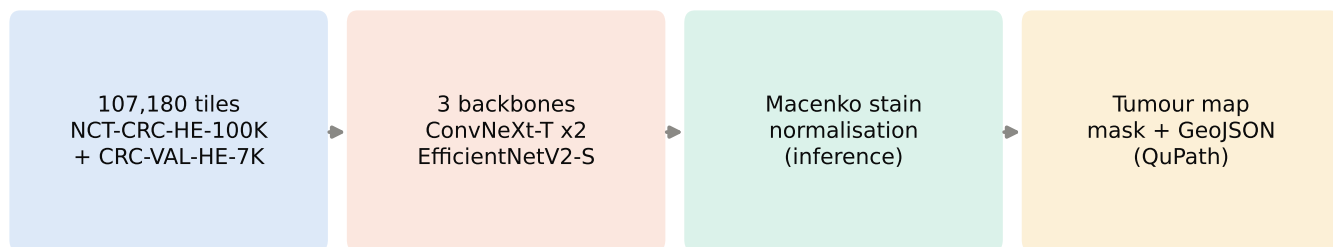
A negative control is necessary but not sufficient, since a model that predicted 'no tumour' everywhere would also pass it. The evidence rests on both sides holding simultaneously, which they do.

Agreement with an independent modality (Fig. 4). For Cancer_P1, 202,182 segmented cells carried transcriptomically assigned identities from the matched Visium HD assay. Projecting these onto the tile grid gave 2,307 tiles with at least 20 cells, of which 859 were molecularly tumour-dominant. The H&E model, which never observes gene expression, agreed with these labels at AUROC 0.9855 (Spearman rho 0.769), with precision 0.935, sensitivity 0.921 and specificity 0.962 at the released threshold. Every model variant was scored the same way and the released ensemble ranked highest (Fig. 4f).

Figure 1. Construction and benchmarking of omicoretumor-crc-he-v0.1

a

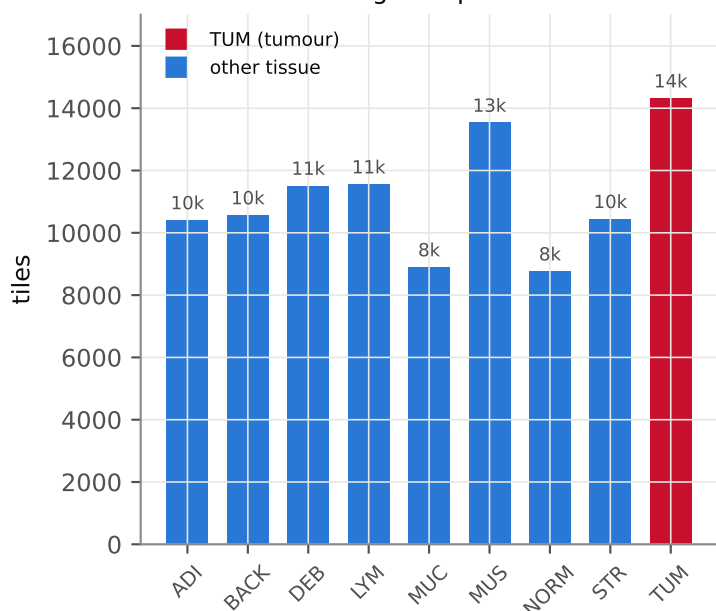
Model construction



9 tissue classes: ADI BACK DEB LYM MUC MUS NORM STR TUM (TUM = adenocarcinoma epithelium)

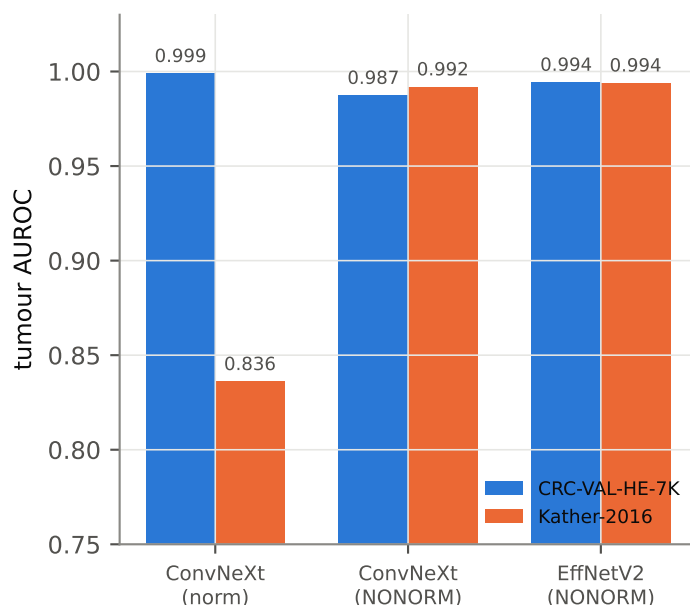
b

Training tiles per class



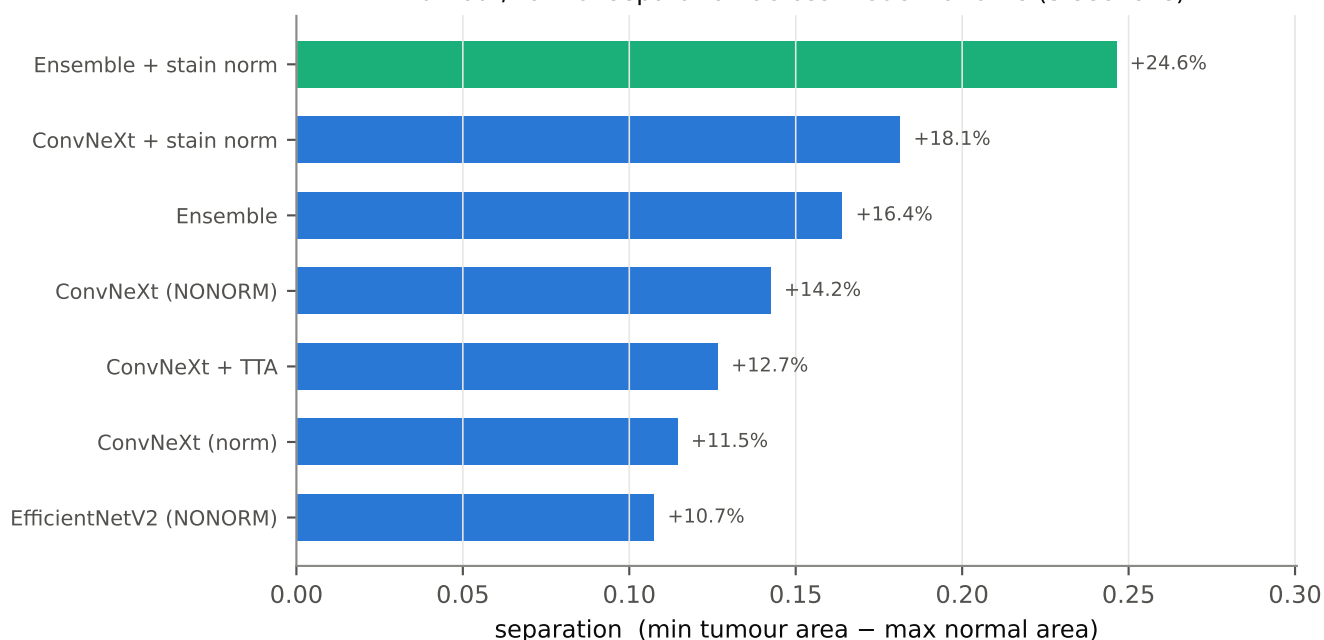
c

Held-out tumour AUROC



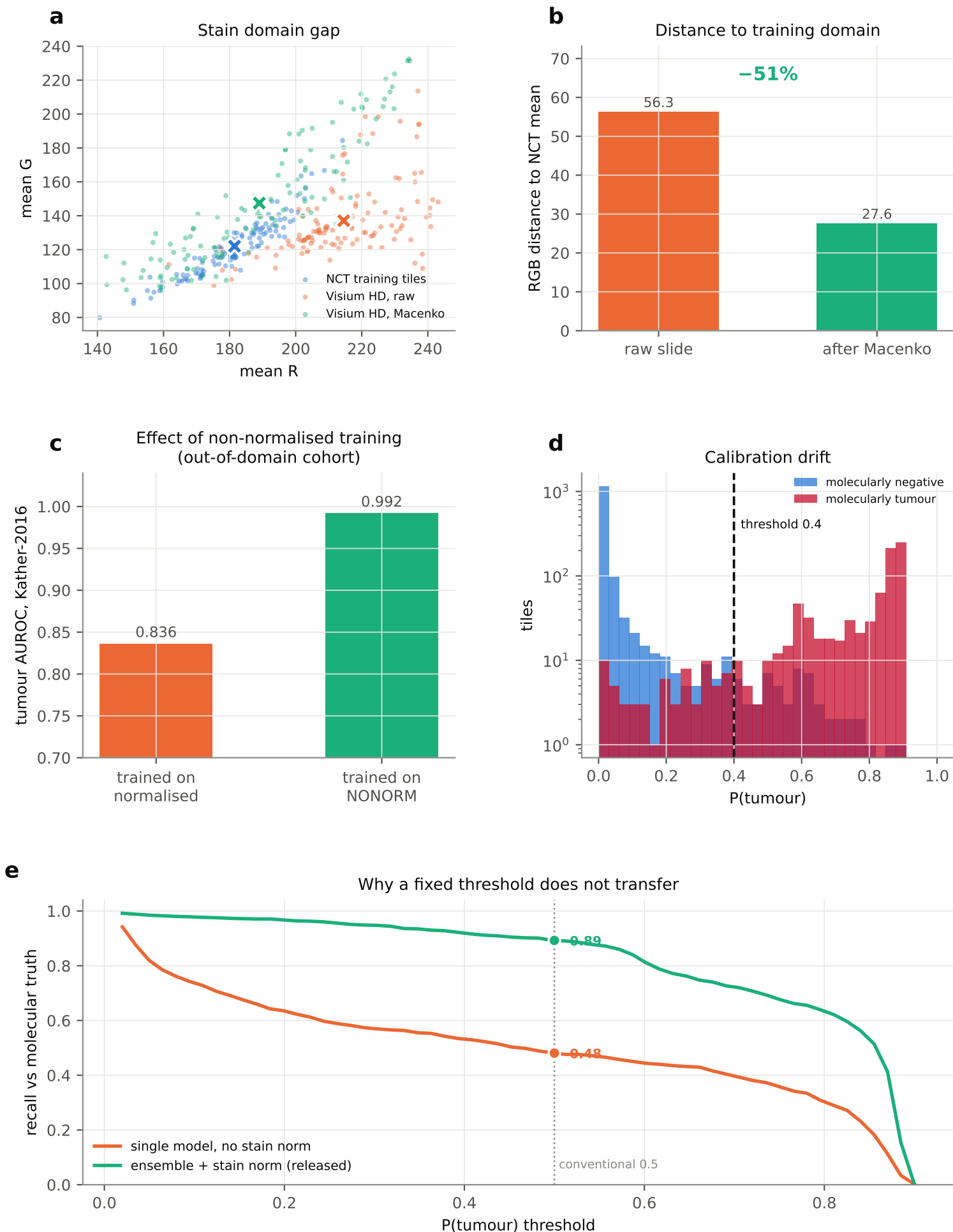
d

Tumour/normal separation across model variants (5 sections)



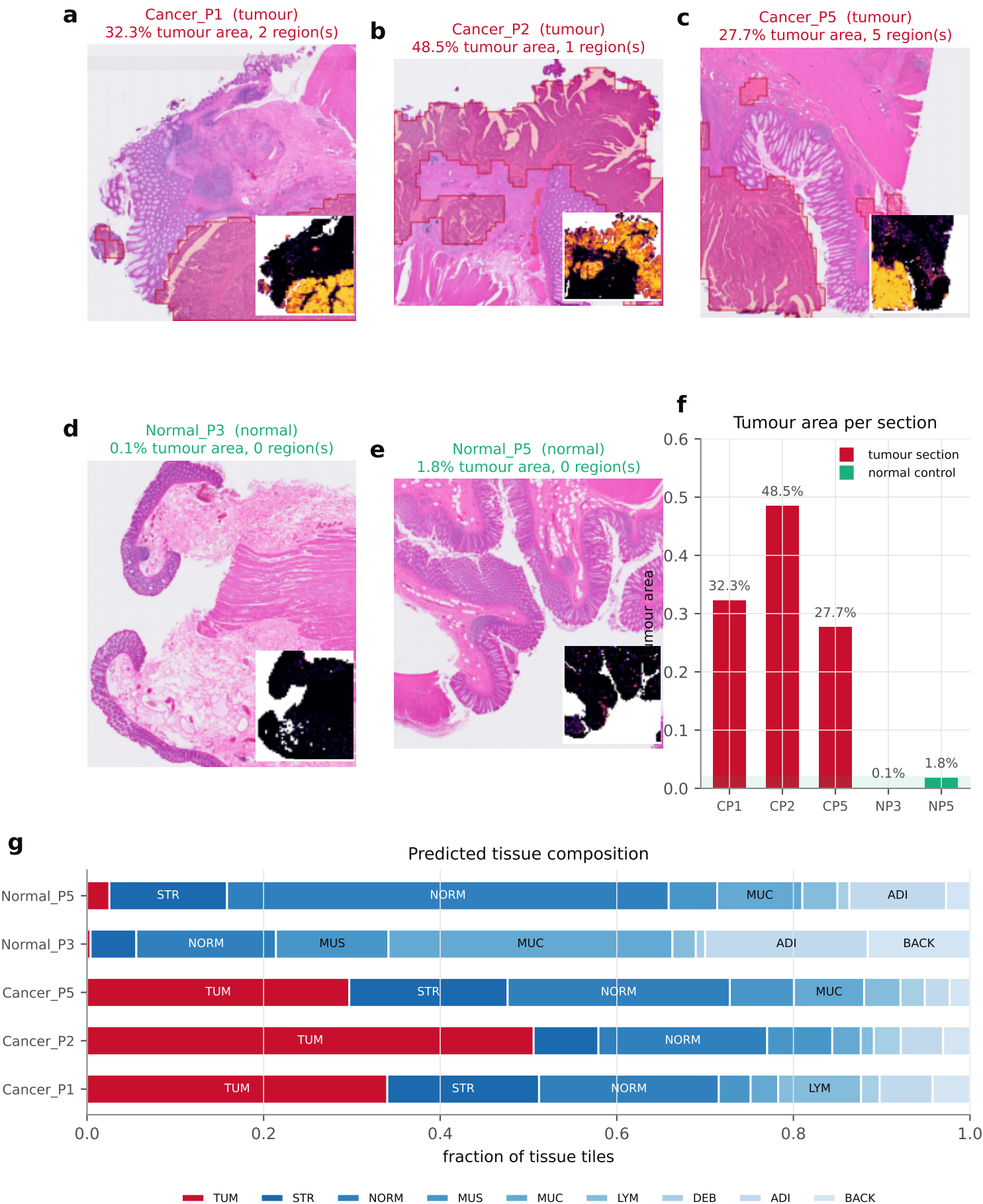
(a) Pipeline. (b) Class composition of the 100,000-tile training set. (c) Tumour AUROC on two held-out cohorts; Kather-2016 is a different scanner and cohort, so it measures cross-domain transfer. (d) Separation on the five Visium HD sections; >0 means tumour and normal sections do not overlap. The ensemble with stain normalisation (green) ranks first.

Figure 2. Stain-domain shift is the dominant failure mode, and how it is mitigated



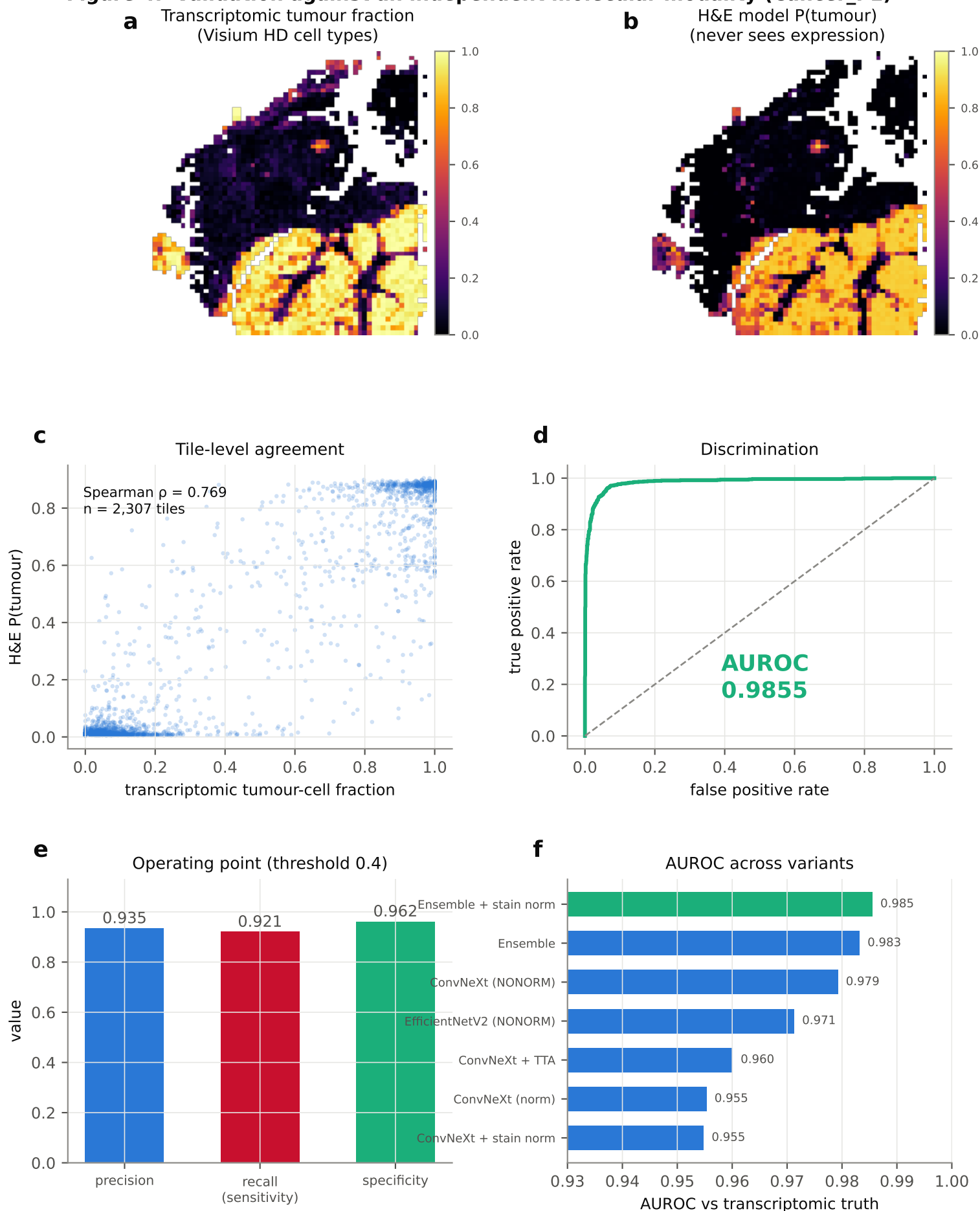
(a,b) Target slides are lighter and pinker than the training tiles; Macenko normalisation halves the gap. (c) Training on non-normalised tiles raises cross-cohort tumour AUROC from 0.836 to 0.992. (d) On target slides the two molecular classes stay well separated but the probabilities compress toward zero. (e) Consequence: at the conventional 0.5 threshold a single model recovers roughly half the tumour it ranks correctly; the released ensemble restores recall. Thresholds must be re-checked on each new scanner.

Figure 3. Application to three colorectal tumours and two matched normal controls



(a-e) H&E with the predicted tumour region outlined and tinted; inset is the per-tile probability map. (f) Tumour sections occupy 27.7-48.5% of tissue while normal controls stay below 2% (shaded band) and export no regions at all. (g) Predicted tissue composition: TUM dominates the tumour sections and is near-absent from the controls, where normal mucosa, muscle and mucus dominate.

Figure 4. Validation against an independent molecular modality (Cancer_P1)



202,182 Visium HD cells were assigned transcriptomic identities and projected onto the tile grid; 2,307 tiles had ≥ 20 cells. The image model never observes gene expression, so agreement is orthogonal evidence rather than a morphology-vs-morphology comparison. (a,b) The two maps localise the same compartment. (c,d) Tile-level agreement and discrimination. (e) At the released threshold. (f) Every variant, ranked; the released ensemble is highest. The 0.4 operating point was selected on this same section, so (e) is optimistic; AUROC is threshold-free.

Methods

Data. All training data is openly licensed under CC-BY-4.0. NCT-CRC-HE-100K provided 100,000 224x224 px tiles at 0.5 um/px across nine classes (ADI, BACK, DEB, LYM, MUC, MUS, NORM, STR, TUM), used in both its stain-normalised and non-normalised releases; CRC-VAL-HE-7K (7,180 tiles, separate cohort) and the Kather-2016 collection (4,375 usable tiles after class mapping) served as held-out test sets. EBHI-SEG supplied 2,228 image/mask pairs for an auxiliary segmentation model. The five evaluation sections are 10x Visium HD colorectal FFPE samples and share no material with any training set.

An important caveat applies to the internal split: NCT-CRC-HE-100K ships no patient identifiers, and the code embedded in each filename is a per-tile hash that we verified to be unique for every tile. A random split therefore probably places tiles from the same patient on both sides, so the internal validation figure is optimistic and we quote only held-out cohorts.

Models. Three classifiers were fine-tuned for 8 epochs with AdamW, one-cycle scheduling, label smoothing 0.1 and class-weighted cross-entropy, in bfloat16: ConvNeXt-Tiny on the normalised tiles, and ConvNeXt-Tiny and EfficientNetV2-S on the non-normalised tiles. Augmentation targeted stain robustness (random resized crop, dihedral flips and 90-degree rotations, colour jitter, random erasing). The released model averages the three softmax outputs. All pretrained backbones are Apache-2.0.

Inference. Images are rescaled so that one pixel is 0.5 um, tiled at 224 px, and tiles whose saturation indicates glass rather than tissue are skipped. Each tile is Macenko-normalised toward a fixed H&E target before classification. The per-tile tumour probability is thresholded at 0.4, cleaned with a morphological close-open, and traced to polygons that are written as GeoJSON in the source image's coordinate frame. A 13,000 x 13,000 px section takes about 50 s on one RTX 5080, including normalisation.

Molecular validation. Visium HD cells were mapped into image coordinates using the recorded bin2cell geometry: image pixels equal the cropped spatial coordinates multiplied by the recorded scale factor, which we confirmed equals the source resolution divided by 0.5 um. The cell-identifier join matched 202,182 of 202,182 cells, and every cell fell inside the scored grid. Tiles with at least 20 cells were scored, a tile counting as molecularly positive when at least half its cells carried a tumour identity.

Reproducibility. Training, evaluation and figure code ship with the package. `run_experiments.sh` reproduces the full matrix - three models, seven inference variants, two held-out benchmarks, the section-level dry test and the molecular comparison - and is resumable. The development log, including approaches we tested and rejected, is in `FINDINGS.md`.

Discussion

The useful finding here is not that a convolutional ensemble can classify colorectal tissue; that has been established for years. It is that the barrier to reuse is calibration rather than capability. A model whose AUROC is unchanged can lose half its sensitivity on a new laboratory's slides purely because the probability scale has shifted, and it will do so quietly, presenting as unusually high precision. Any group deploying a tile classifier on its own material should verify the operating point before trusting the output, and we ship the tools to do so rather than only a threshold.

We deliberately did not adopt two techniques that appeared promising. Spatial smoothing of the probability map, motivated by the contiguity of tumour, improved AUROC by 0.002 while reducing recall in every configuration tested. Label-free automatic thresholding recovered recall on tumour-bearing slides but fabricated disease on tumour-free tissue, calling 39% of a lung section colorectal tumour; it is retained only behind an explicit opt-in. Both are documented with their numbers so that others need not repeat them.

Limitations

The evaluation is small. Three tumour and two normal sections from a single cohort demonstrate feasibility, not generalisation, and matched transcriptomics were available for one section only. The reported precision and sensitivity at threshold 0.4 are optimistic because that threshold was chosen on the same section; the AUROC of 0.9855 is threshold-free and unaffected.

The model marks regions, not cells. Boundaries are resolved at 224 px, or 112 um, so a called region necessarily contains stroma, immune and vascular cells alongside malignant glands, and small infiltrating glands at a margin may be missed. Claims of the form 'identifies malignant cells' are not supported by this evidence; claims of the form 'localises tumour-enriched regions' are.

The model is colorectal-specific. Lung and prostate sections were used as out-of-organ negative controls and produced 8.9% and 1.7% spurious tumour area at the released threshold. Adenoma, dysplasia, inflammation, ulceration, necrosis, mucinous variants and treatment-altered tissue were not represented in evaluation, and a hard-negative analysis on such tissue is the natural next step.

What would strengthen this

In order of value: independent pathologist annotation of invasive carcinoma on the same sections, scored with Dice and boundary distance rather than tile metrics; molecular validation extended to the remaining four sections, which requires only re-running an existing annotation pipeline; and external validation across 20-50 unseen patients spanning scanners and histological variants, including difficult non-neoplastic tissue.

Data and code availability

Package	pip install omicoretumor		
Source	github.com/OmiCore/omicoretumor (Apache-2.0)		
Model	omicoretumor-crc-he-v0.1		
Training	NCT-CRC-HE-100K	doi:10.5281/zenodo.1214456	CC-BY-4.0
	Kather-2016	doi:10.5281/zenodo.53169	CC-BY-4.0
	EBHI-SEG	doi:10.6084/m9.figshare.21540159	CC-BY-4.0
Evaluation	10x Visium HD colorectal FFPE (Oliveira et al. 2025)		

References

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